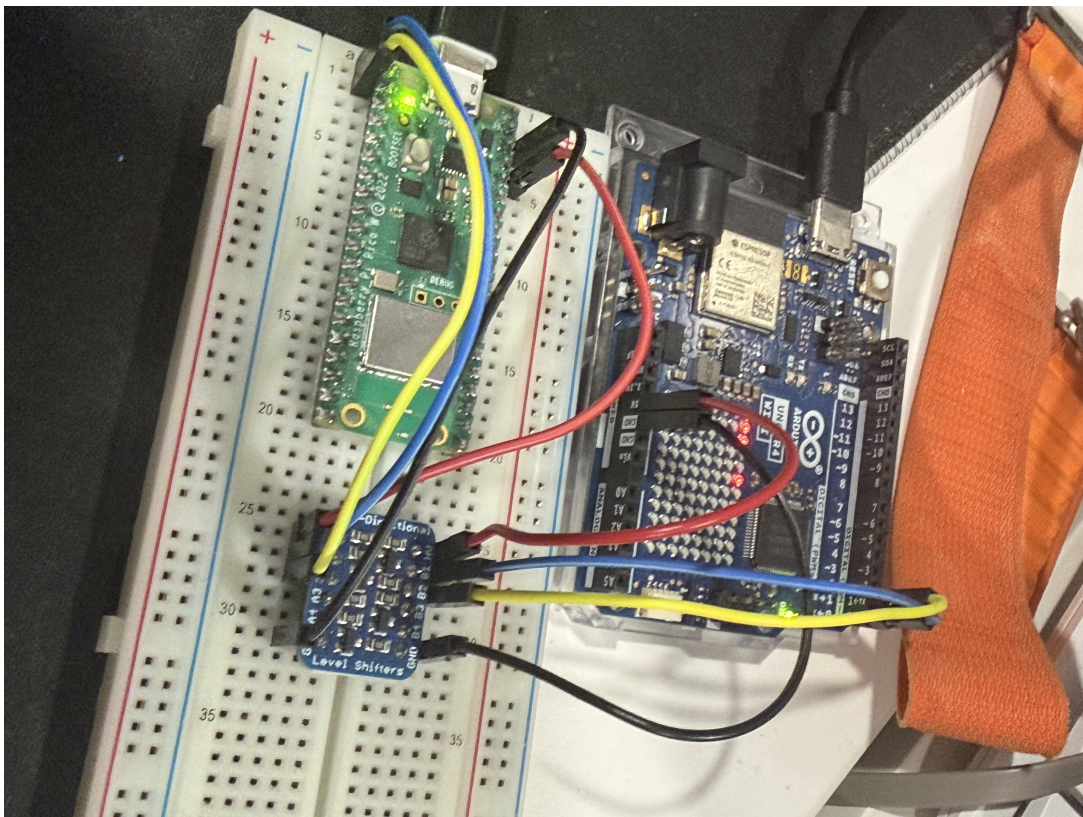


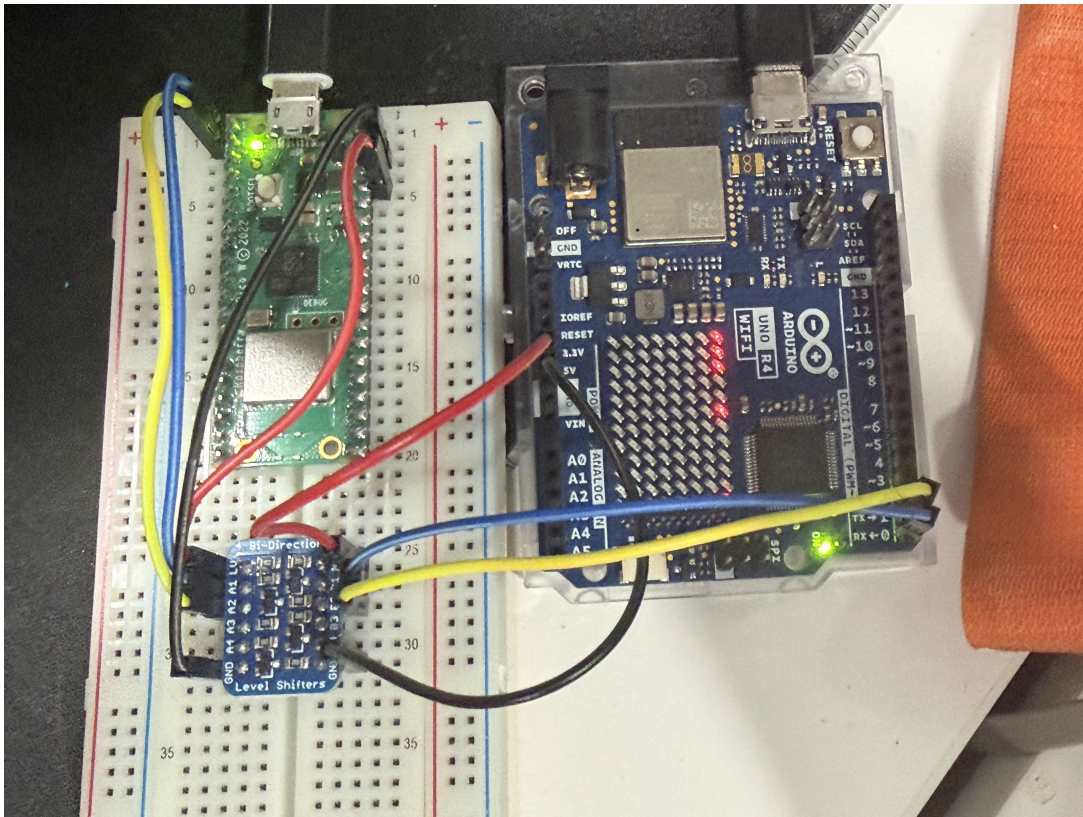
Connection guide: photographed build and proposed audio output

Status (2026-09-23): The board and jumper layout below is photographed, and the owner confirms the return-wire level shifter and separate USB power. The Uno `A0` audio output is **not yet connected**. The line-output circuit on this page is a **proposed, unbuilt design** for a powered speaker's line input or mixer, not a measured property of the current instrument. The firmware snapshot is [2026-09-23](#).

The current setup



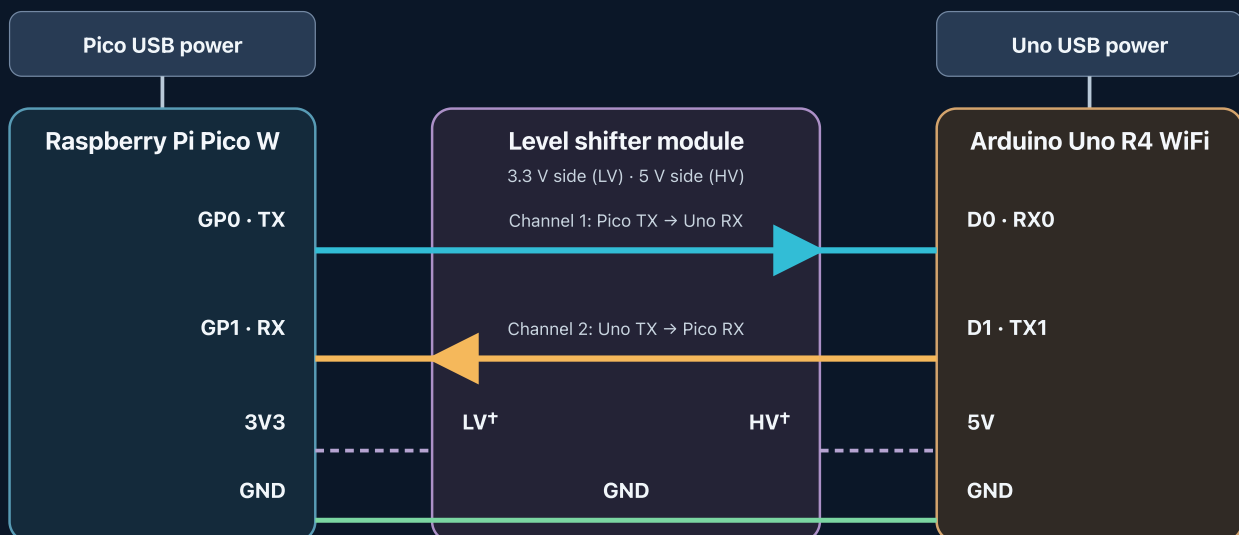
Owner-supplied photograph, 2026-09-23. Image metadata removed for the guide. The boards are powered through their individual USB ports; audio is not wired.



The lower-right Uno `RX0` / `TX1` header, Pico, and level-shifter module are visible. Wire colors alone do not establish an electrical pin map or voltage.

Pico W ↔ Uno R4 WiFi

Current USB-powered MIDI UART topology · 31,250 baud · signal colors show direction, not jumper colors



[†] Shifter supply terminal labels and exact channel pin pairs still need a continuity check.

The two USB 5 V rails are not joined. Audio, DIN connector, and headphone circuitry are not installed here.

Current connection schematic based on the installed firmware, the photographs, and the owner's confirmation. Its blue and amber lines indicate MIDI direction, **not jumper colors**. It records the present two-board topology, not the proposed audio circuit below. The exact level-shifter channel numbers and LV/HV supply-pin endpoints have not been continuity-checked.

In the photographs, both UART signal jumpers appear to enter separate channels of the level-shifter module. The owner has specifically confirmed **Uno TX → 5 V-to-3.3 V shifting → Pico RX** and a common ground. The Pico-to-Uno signal path is defined by the two sketches, but the module's individual channel pins are inferred from the photograph. The dashed shifter-supply lines show the expected Pico 3.3 V/LV and Uno 5 V/HV sides; check the actual terminal labels and continuity before treating them as a measured netlist. The boards remain independently USB-powered; their 5 V rails are not tied together.

Path	Firmware or owner-confirmed connection	Status
Power	Pico W and Uno R4 WiFi each use their own USB port.	Owner-confirmed. Exact USB supply arrangement and voltages have not been measured. Do not join the boards' 5 V rails for this setup.
Common reference	Pico and Uno grounds are connected through the level-shifter wiring.	Owner-confirmed; the photos show ground wires but cannot prove continuity.
Pico → Uno	Pico <code>Serial1</code> TX, GPIO 0 → a level-shifter channel on the Pico 3.3 V side → Uno <code>Serial1</code> RX (<code>rx0</code> /D0), 31,250 baud MIDI UART.	Source-defined path; photographed shifter routing inferred. Exact channel and endpoints have not been continuity-checked.
Uno → Pico	Uno <code>Serial1</code> TX (<code>tx1</code> /D1) → a separate 5 V-to-3.3 V level-shifter channel → Pico <code>Serial1</code> RX, GPIO 1.	Owner-confirmed path and shifter; exact module pins and Pico RX voltage still need measurement.
Browser ↔ Pico	Pico serves the panel by Wi-Fi; browser control uses WebSocket on port <code>8081</code> .	Source-defined; deployment smoke check showed both connection labels.
Separate DIN MIDI	Uno software serial RX is D2; TX is configured on D3.	Source-defined only. A DIN connector and its input circuit are not identified in these photographs.
Audio	Uno writes 12-bit samples to DAC pin <code>A0</code> at a nominal 22,050 samples/s.	Not built: no A0 output circuit, jack, or powered-speaker/mixer connection yet.

Pico UART setup, Uno MIDI instances and DAC output, technical guide.

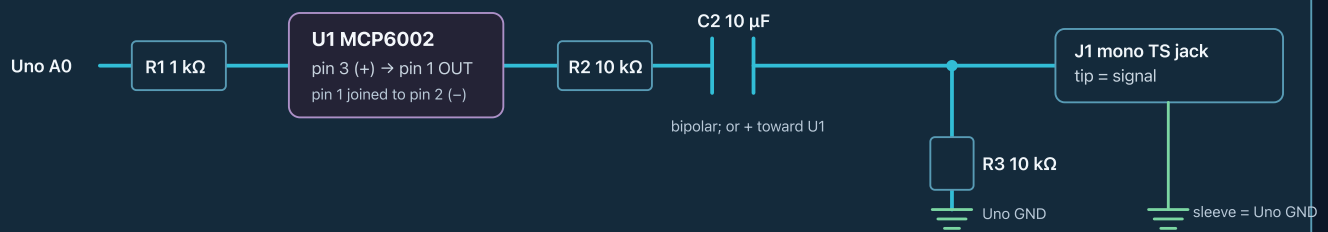
Proposed mono line output from Uno A0

This circuit makes a **mono, AC-coupled line output** for a powered speaker **line input** or mixer line input. It uses a high-input-impedance buffer so the Uno's DAC is not loaded directly by the cable or destination. The 10 kΩ/10 kΩ output network reduces the possible signal level before it reaches the destination. It is **not** a speaker driver or headphone output. The diagram separates its audio, power, and unused-amplifier connections so each path can be read without crossing another.

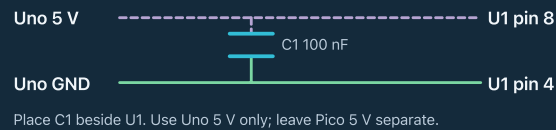
Uno R4 A0 → buffered mono line output

One path for audio, one for power. Proposed circuit; it has not been built or measured.

1 Audio signal: read left to right



2 Power the buffer from the Uno



3 Tie off the unused half of U1

Pin 5 (+B) → Uno GND
Pin 6 (-B) → pin 7 (OUT B)
All ground symbols on this page connect to Uno GND.

Diagram colors indicate signal, supply, and ground functions; they do not identify jumper colors.

What is U1? The **MCP6002** is a **separate, small analog chip** to add to the breadboard; it is not already inside the Uno, Pico, or TDA1308 module. It has two amplifiers in one package. This circuit uses one as a unity-gain **buffer**: the sound voltage stays approximately the same, while the following line and headphone circuits draw their signal from U1 instead of loading the Uno DAC. For a breadboard, look for **MCP6002-I/P**, the 8-pin through-hole PDIP version shown in [Microchip's datasheet](#). An SOIC or MSOP version needs an adapter and a different physical mounting method. The chip itself needs no programming.

Part	Value and connection
U1	MCP6002 dual, rail-to-rail op amp, DIP-8 pinout shown . Power pin 8 from Uno 5 V and pin 4 from Uno GND . The op amp uses the Uno's USB-derived 5 V only; leave the Pico 5 V rail separate.
C1	100 nF ceramic directly across U1 pins 8 and 4, close to the IC.
R1	1 kΩ from Uno A0 to U1 pin 3 (+A). U1 pin 2 (-A) joins pin 1 (OUT A) for unity-gain buffering.
Unused U1 channel	Join pins 6 (-B) and 7 (OUT B); connect pin 5 (+B) to Uno GND. Do not leave its inputs floating.
R2	10 kΩ from U1 pin 1 to coupling capacitor C2.
C2	10 μF coupling capacitor. A bipolar/nonpolar part is simplest. For a polarized electrolytic, its positive lead faces U1/R2 and its negative lead faces the jack.
R3	10 kΩ from the output side of C2 to Uno GND; provides a DC reference and sets the nominal output divider with R2.
J1	Mono 6.35 mm or 3.5 mm TS line jack: tip to the output side of C2, sleeve to Uno GND. Use a short shielded cable to the destination's line input .

The Uno sketch centers its signed audio at DAC midscale before `analogWrite(A0, ...)`; C2 removes that DC component before the line jack. The [Arduino board documentation](#) identifies A0 as its 12-bit DAC. The [Renesas RA4M1 datasheet](#) specifies a minimum 30 k Ω resistive DAC load and maximum 50 pF capacitive load in its stated test conditions, which is why the circuit buffers A0 rather than attaching a jack, headphones, or a filter capacitor to it. The [MCP6002 product page and datasheet](#) cover the proposed 5 V, unity-gain buffer; check the package pinout before wiring another op amp or package.

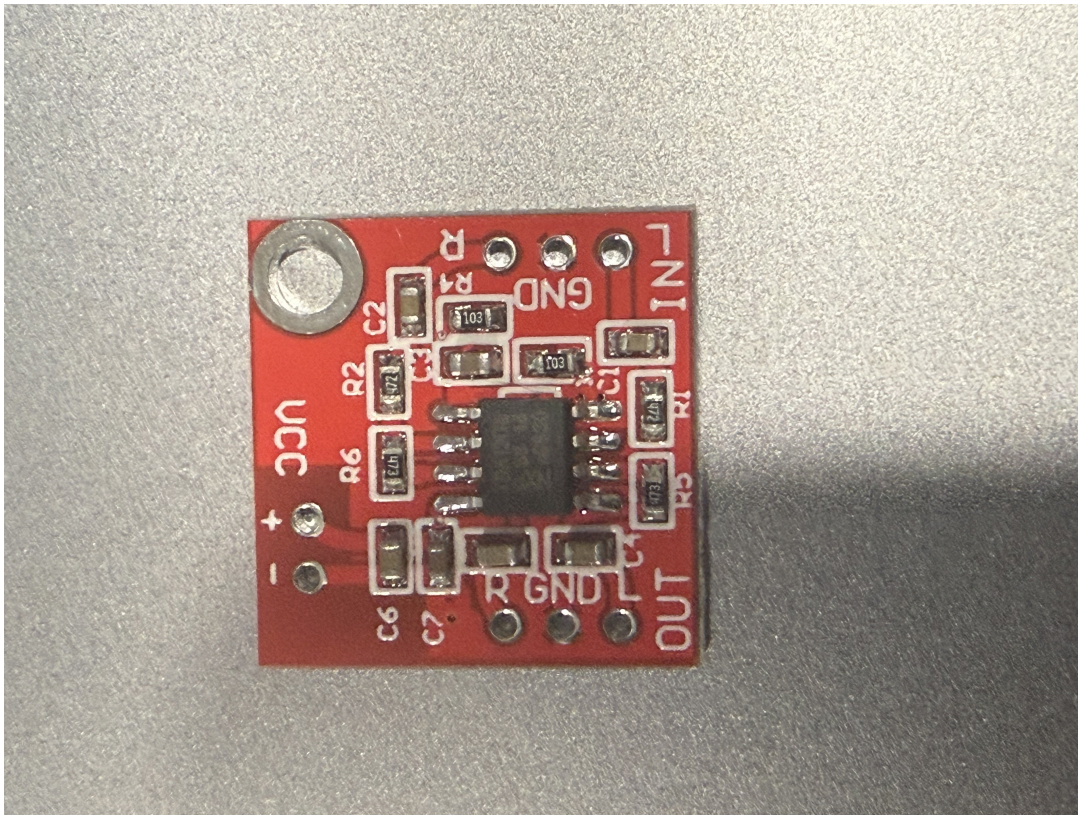
Build and first-power checks

1. Unplug both USB cables. Build U1, C1, R1, R2, C2, R3, and J1 as shown. Connect the new circuit's 5 V and ground **only to the Uno**. Keep the existing Pico–Uno ground link and level shifter as they are.
2. Before plugging into a mixer or powered speaker, check continuity: jack sleeve $\rightarrow$ Uno GND, U1 pin 8 $\rightarrow$ Uno 5 V, U1 pin 4 $\rightarrow$ Uno GND. Confirm no 5 V-to-GND short and verify C2 polarity if it is electrolytic.
3. Power the Uno by USB. With a multimeter, check approximately 5 V between U1 pins 8 and 4. Check the jack tip against sleeve for **near-zero DC** after C2 settles. If a substantial DC voltage persists, disconnect the audio destination and correct the circuit.
4. Set the destination input gain and speaker level low. Connect the mono line output to a line input, play a note, then raise the listening level gradually. Record hum, clipping, and measured jack levels in the [hardware test log](#).

The divider gives approximately one-half of the buffered AC voltage into a high-impedance input; a 10 k Ω mixer input gives approximately one-third. These are circuit estimates, **not measured QuarkWave output levels**. A stereo aux input needs a separate mono-to-stereo distribution arrangement; do not tie its left and right outputs or contacts together. A passive loudspeaker or headphones need an amplifier.

Proposed headphones with the owner's TDA1308 board

The owner identified the pictured module from its [purchase listing](#) as a **TDA1308 headphone amplifier board**. Its photographed face has VCC +/–, IN L/GND/R, and OUT L/G/R pads. The chip marking and the board's component values cannot be read from the photo, so its internal circuit and actual gain remain unverified. The [NXP TDA1308 datasheet](#) describes a stereo headphone driver that operates from a single 3–7 V supply and is characterized with 32 Ω headphones; that supports 5 V as a chip supply but does not by itself validate this particular assembled module.

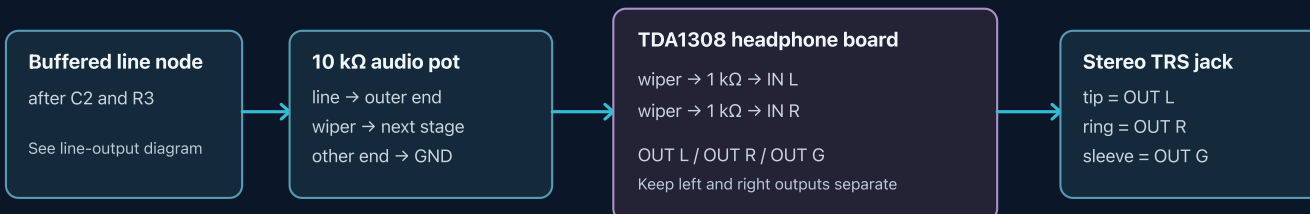


Owner-supplied photograph, 2026-09-23; metadata removed. The printed `IN`, `OUT`, and `VCC` labels identify the pad groups. The module is not connected to QuarkWave yet.

Use the **buffered and AC-coupled output node** shown after C2/R3 in the line-output diagram as the headphone board's source. It is mono; feed that one source to *both inputs*, but keep the board's left and right **outputs separate**. Add a physical input volume control before the headphone module; the Uno's master gain cannot turn fully to zero.

Mono sound → both ears, with a separate volume control

Proposed connection map. The TDA1308 board has not yet been wired or tested.



Power and common reference

Uno 5 V → board VCC + • Uno GND → board VCC – and IN GND • Verify OUT G → Uno GND continuity

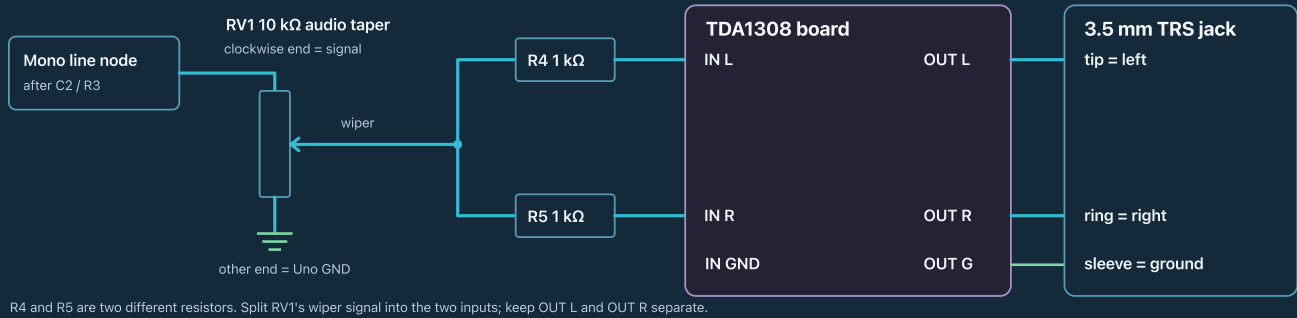
Diagram colors indicate function, not jumper colors. Module internals and output levels remain unverified.

The map above shows the order of the modules. This second drawing shows the **individual external components and connections**. It starts at the same output node after C2/R3 in the line-output schematic; it does not connect directly to Uno A0. The TDA1308 board is shown as a labeled module because its internal PCB schematic has not been verified.

Buffered mono signal → TDA1308 → stereo headphones

External wiring schematic. The TDA1308 board's internal components are not shown or verified.

1 Audio and volume



2 Supply and ground connections

Uno 5 V → board VCC +

Uno GND → board VCC – and IN GND

Check OUT G ↔ Uno GND continuity

Before inserting headphones: volume fully down, then measure OUT L and OUT R for near-zero DC relative to OUT G.

Diagram colors indicate function, not jumper colors. Confirm the board pad labels before wiring.

From	To	Purpose
Output node after C2/R3	One outer terminal of a 10 kΩ audio-taper potentiometer	Mono headphone level control. The other outer terminal goes to Uno GND.
Potentiometer wiper	Two separate 1 kΩ resistors , then module IN L and IN R respectively	Feeds the mono sound to both headphone channels without joining their outputs.
Uno GND	Module IN GND and VCC –	Common input and supply reference. Confirm module OUT G is continuous with this ground before fitting the jack.
Uno 5 V	Module VCC +	Module power from the Uno's USB-derived rail; do not connect Pico 5 V here.
Module OUT L / OUT R / OUT G	3.5 mm stereo TRS jack tip / ring / sleeve	Left ear / right ear / common return. Do not join OUT L and OUT R .

First check: Leave headphones unplugged, set the physical volume control fully down, power the Uno, and measure DC from each module output (**OUT L**, **OUT R**) to **OUT G**. Both should settle near zero before headphones are connected. If either output holds substantial DC, stop and inspect the board and wiring. Then test with inexpensive headphones at minimum level and raise the control slowly. The board's headphone response, output coupling, and possible USB noise remain **hardware untested**. This route does not replace the mono line output for a mixer or powered speakers.

Still to establish on the assembled build

Item	Evidence needed
Pico RX logic level	Measure Uno return-data high level at Pico GPIO 1 and confirm it stays within 3.3 V logic limits.
UART and common-ground pin map	Continuity-check both UART channel pairs and the photographed jumper endpoints; record the shifter's actual HV/LV supply and ground terminals.
USB supply and noise	Record the two USB power sources and measure supply voltage and any audio hum once the line stage exists.
Proposed line stage	Build, photograph, measure DC and audio at the jack, and test with the intended powered speaker or mixer.
TDA1308 headphone module	Build the buffered feed and volume control, confirm the pad ground with continuity, measure output DC with no headphones, then test low-volume playback.
DIN MIDI input	Provide the connector and input-interface circuit if external DIN operation is part of this build.

The [quick start](#) continues to assume a listening setup supplied by the builder until the line-output circuit is assembled and verified.